

# Andrea Brilli

✉ brilli@diag.uniroma1.it

📍 Sapienza, University of Rome, Department of Computer, Control and Management Engineering "Antonio Ruberti"

🌐 brilliandrea    🔗 andrea-brilli-161934199

## Education

|                                                                                                                                                                                                                                                                                           |             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>Sapienza, University of Rome</b> , Operations Research                                                                                                                                                                                                                                 | Rome, Italy |
| <ul style="list-style-type: none"><li>Curriculum: Operations Research</li><li>Thesis: Derivative-Free Optimization: worst-case complexity for Line-Search methods and a mixed Penalty-Barrier approach</li><li>ABRO (Automatic control, Bioengineering and Operations Research)</li></ul> | 2021 – 2025 |
| <b>Sapienza, University of Rome</b> , Management Engineering                                                                                                                                                                                                                              | Rome, Italy |
| <ul style="list-style-type: none"><li>Curriculum: Decision Models</li><li>Thesis: An interior penalty method for black-box constrained optimization</li></ul>                                                                                                                             | 2019 – 2021 |
| <b>Sapienza, University of Rome</b> , Management Engineering                                                                                                                                                                                                                              | Rome, Italy |
| <ul style="list-style-type: none"><li>empty</li></ul>                                                                                                                                                                                                                                     | 2015 – 2019 |

## Experience

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <b>Sapienza University of Rome</b> , PostDoc                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Rome, Italy      |
| Department of Computer, Control and Management Engineering. Funded by PRIN (NextGenerationEU funds) for hydrodynamic optimization study on biomimetic and innovative hydrodynamic configurations for autonomous underwater gliders (AUG), specifically designed for ocean surveying.                                                                                                                                                                                                                                                                                                                                                                          | 2024 – present   |
| <ul style="list-style-type: none"><li>Support to PhD students on advanced techniques for training neural networks and derivative-free techniques</li><li>Collaboration with INESC MN (Lisbon, Portugal) for optimization of structural design parameters of metamaterials</li><li>Collaboration with NOVA University of Lisbon on techniques to approximate the Pareto Front of constrained multiobjective optimization problems</li><li>Collaboration with Polytechnique Montréal to study nonsmooth constrained optimization problems</li><li>Collaboration with University of Florence to enhance performance of Augmented Lagrangian techniques</li></ul> | 2 years          |
| <b>European Space Agency - ESRIN - <math>\phi</math>-lab</b> , Research Visiting                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | ESRIN, Italy     |
| Collaboration to develop denoising techniques for satellite data striping noise.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2026 – present   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1 year           |
| <b>Polytechnique Montréal</b> , Ph.D. Visiting Researcher                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Montréal, Canada |
| Six months of collaboration with Professor Sébastien Le Digabel and Professor Youssef Diouane to enhance the performance of NOMAD solver.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2024 – 2024      |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1 year           |
| <b>Universidade NOVA de Lisboa</b> , Ph.D. Visiting Researcher                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Lisbon, Portugal |
| One year of collaboration with Professor Ana Luísa Custódio and Ph.D. Everton Jose da Silva to enhance the performance of the SID-PSM algorithm.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2022 – 2023      |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1 year           |

## Teaching

|                                                                                                                                                                             |                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>Sapienza University of Rome</b> , Mathematical Programming                                                                                                               | Rome, Italy          |
| <ul style="list-style-type: none"><li>Introduction to multivariate calculus and mathematical optimization</li><li>Bachelor degree in Computer Science Engineering</li></ul> | Sept 2025 – Jan 2026 |

## Awards

---

### AIROYoung Dissertation Award

Sept 2025

Yearly award for the best doctoral thesis in Operations Research defended in Italy.  
AIROYoung (Italian Operations Research Society)

## Publications

---

### Complexity results and active-set identification of a derivative-free method for bound-constrained problems

Accepted for publication in JOTA. Analysis of complexity and active-set identification for derivative-free optimization with bound constraints.

A. Brilli, A. Cristofari, G. Liuzzi, S. Lucidi

[arxiv.org/abs/2402.10801](https://arxiv.org/abs/2402.10801)

### A penalty-interior point method combined with mads for equality and inequality constrained optimization

Novel constraint-handling method combining penalty and interior point approaches with MADS algorithm.

C. Audet, A. Brilli, Y. Diouane, S. Le Digabel, E. Silva, C. Tribes

[arxiv.org/abs/2601.20811](https://arxiv.org/abs/2601.20811)

### Tunable ito-metal plasmonic metamaterial channel for tailored sensing: A simulation-driven approach

Optimization-driven design of plasmonic metamaterial sensors for tailored sensing applications.

C. Alfisi, A. Brilli, H. Terças, S. Cardoso

[doi.org/10.1063/5.0297834](https://doi.org/10.1063/5.0297834)

### An interior point method for nonlinear constrained derivative-free optimization

Novel interior point approach for solving general nonlinear constrained problems without derivatives.

A. Brilli, G. Liuzzi, S. Lucidi

[doi.org/10.1080/10556788.2025.2453110](https://doi.org/10.1080/10556788.2025.2453110)

### Nonlinear derivative-free constrained optimization with a mixed penalty-logarithmic barrier approach and direct search

Mixed penalty-barrier method combined with direct search for general constrained derivative-free optimization.

A. Brilli, A. L. Custódio, G. Liuzzi, E. J. Silva

[arxiv.org/abs/2407.21634](https://arxiv.org/abs/2407.21634)

### Worst case complexity bounds for linesearch-type derivative-free algorithms

Theoretical complexity analysis for linesearch-based derivative-free optimization methods.

A. Brilli, M. Kimiaiei, G. Liuzzi, S. Lucidi

[doi.org/10.1007/s10957-024-02519-x](https://doi.org/10.1007/s10957-024-02519-x)

## Skills

---

**Optimization**

**Programming**

**Software Development**

## Languages

---

**Italian**

Native speaker

**English**

Fluent

**Portuguese**

Basic

**Spanish**

Basic

## Certificates

---

|                                                                                                                            |      |
|----------------------------------------------------------------------------------------------------------------------------|------|
| <b>Deep Learning Specialization</b>                                                                                        | 2020 |
| <b>Data-driven Astronomy</b>                                                                                               | 2020 |
| <b>CyberChallenge 2020</b><br>Training programme organized by CINI in collaboration with the Ministry of Defence and SISR. | 2020 |

## Projects

---

|                                                                                                                                                   |                |
|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <b>DFL - Derivative-Free Library</b><br>Collaborator for the implementation of algorithms in Python within the DFL library.<br>Author of LOG-DFL. | 2021 – present |
| <b>LOG-DS</b><br>Co-author of the LOG-DS algorithm, a direct search method for general nonlinear constrained problems implemented in Matlab.      | 2023 – present |
| <b>NOMAD</b><br>A Blackbox Optimization Software. Supported the implementation of new constraint-handling strategies within the package.          | 2024 – present |

## References

---

**Professor Giampaolo Liuzzi****Professor Stefano Lucidi****Professor Ana Luísa Custódio****Professor Sébastien Le Digabel****Professor Youssef Diouane**